Data Attribution and Sources

Primary Data

• V-Dem (Varieties of Democracy) Country-Year Full+Others, version 15 (2024 data release). Released component fields: v2cltrnslw, v2juhcind, v2juncind, v2clkill, v2cltort, v2xcl_acjst, and v2x_freexp. The loader also reads v2x_clphy and v2clrspct, but the released 2026 component table reuses v2clkill for lever 6 and v2xcl_acjst for lever 7.

• World Bank World Development Indicators (WDI) via API — GDP per capita (current US$), population, prevalence of undernourishment, and poverty headcount at $2.15 (2017 PPP). Values are the most recent published as of atlas production (labeled 2026 baseline).

• United Nations Member States list (193 sovereign states) for country scope and ISO3 codes.

Source Framework

The RTLDI equation, the nine RTLP indicators, and the nested causal interpretation are taken from:

Sid J.A. Hubbard, Causality and Attraction: A Continuum of Steady States (Version 3, May 2026), DOI: 10.5281/zenodo.19468550. The operational crosswalk and binarization thresholds used here are documented in docs/indicator_crosswalk.md of this project.

This PDF was generated programmatically from the open RTLDI ATLAS toolkit. No proprietary or restricted data were used.

RTLDI Atlas Suite and Public Repository

This PDF atlas was produced by the RTLDI Atlas Suite, the open repository that contains the canonical data crosswalks, Python build pipeline, choropleth map generation, PDF ebook generator, and self-contained HTML atlas generator used for the public release.

Repository: https://github.com/SJAH9/rtldi-atlas

Users of this atlas are invited to download the RTLDI Atlas Suite and reproduce the published RTLP scores and RTLDI capital-exclusion projections directly from the public-release V-Dem and World Bank datasets. The suite is designed so NGOs, researchers, policymakers, analysts, and investors can rerun the atlas as new public data releases become available.

The same tools can also be used to generate an RTLDI Atlas from proprietary, internal, or field data. The implementation separates the conceptual nine-lever RTLDI model from the data crosswalk: if an organization has its own trusted country, regional, sectoral, or subnational indicators, it can map those fields into the required nine binary RTLP components and then use the existing suite to compute R, per-capita ΔG, total capital exclusions, lever-level attribution, regional summaries, maps, spreadsheets, PDF reports, and HTML outputs.

For public-data runs, use the documented V-Dem and World Bank workflow. For proprietary-data runs, implement a clear data crosswalk, preserve the nine RTLP component meanings, document all thresholds, and keep the output labels tied to the same canonical RTLDI equation. This keeps forks comparable while allowing local datasets to improve coverage, timeliness, or granularity.

The official public atlas has one canonical pipeline: src.build_atlas produces the master atlas table; src.generate_atlas_ebook.prepare_atlas_data() prepares the shared print/web data; and the PDF/HTML generators consume that same prepared data. Forks should preserve this single-pipeline discipline or clearly identify themselves as experimental derivatives.

Nested Causal Modelling

The atlas takes the RTLDI from the source document and extends it via population-weighted regression (25% cap) to quantify capital exclusions — the lost potential GDP where capital is excluded from the economy because the nine protections required for it to operate are not present. The nine levers are the conditions from the source modelling. Their descriptions and the scale of exclusions associated with each are given in the front-matter section. The 25 % cap is a practical limiter on projections; the levers are low-cost conditions whose activation can bring excluded capital back in.

The Nine Levers: Executable Crosswalk

This section mirrors the web atlas lever pages. It reports the global annual capital exclusions attributed to each lever and the exact fields and thresholds used by the executable 2026 pipeline. The lever amounts allocate each country's bounded total across its missing components; they are not nine independent causal estimates and should not be summed without the shared cap.

WHAT IS THE NCM ISOLATED ENTITY?

NCM Isolated Entity is not a V-Dem variable. In this atlas it names the focal unit of analysis: one UN member state observed for one reference year, temporarily treated as a bounded entity so its nine conditions and GDP can be compared consistently. The country is not assumed to be causally isolated; trade, geography, resources, history, conflict, and regional systems remain outside this reduced cross-section.

LEVER 1 OF 9

Legal Protections

$2.78T global annual exclusions

Present when ordinary law is applied consistently to state agents as well as everyone else. Absent when power can act without predictable legal constraint. Multiplies GDP by lowering the risk of arbitrary loss; people and businesses invest and trade more when they know the rules will be enforced fairly. Annual cost is low.

RTLP QUESTION

Does the entity have provisions explicitly protecting the right to life?

FIELDS AND SCORING RULE

Source: V-Dem
Field(s): v2cltrnslw
Binary rule: v2cltrnslw >= 2.0
Operational proxy: Transparent laws with predictable enforcement

Transparency note: Operational de-facto proxy; it is not a direct constitutional right-to-life field.

LEVER 2 OF 9

Independent Judiciary

$2.83T global annual exclusions

Present when there is a judiciary capable of upholding right-to-life laws even against the state. These protections represent opportunities for property rights and contracts to support reliable capital flows to productive uses. Annual cost is low relative to the scale unlocked.

RTLP QUESTION

Is there an independent judiciary capable of upholding right-to-life laws?

FIELDS AND SCORING RULE

Source: V-Dem
Field(s): v2juhcind, v2juncind
Binary rule: mean(v2juhcind, v2juncind) >= 2.0
Operational proxy: High-court and lower-court independence

Transparency note: The two available court estimates are averaged when either is present.

LEVER 3 OF 9

Law Enforcement Accountability

$1.36T global annual exclusions

Present when law enforcement agencies are accountable for unlawful killings and violence. These protections represent opportunities to support greater economic scale by reducing arbitrary risk that can deter investment, labor mobility, and everyday economic activity. Annual cost is primarily institutional oversight.

RTLP QUESTION

Are law-enforcement agencies accountable for unlawful killings?

FIELDS AND SCORING RULE

Source: V-Dem
Field(s): v2clkill
Binary rule: v2clkill >= 2.0
Operational proxy: Freedom from political killings

Transparency note: Outcome/accountability proxy; the same field is reused by lever 6 in the processed 2026 data.

LEVER 4 OF 9

Protection Against Arbitrary Detention

$976.7B global annual exclusions

Present when arbitrary detention is prohibited with real legal recourse. These protections represent opportunities for individuals and families to plan, work, and invest with greater security, supporting scale in the economy. Annual cost is low.

RTLP QUESTION

Is arbitrary detention prohibited with legal recourse?

FIELDS AND SCORING RULE

Source: V-Dem
Field(s): v2xcl_acjst
Binary rule: v2xcl_acjst >= 0.5
Operational proxy: Access to justice index

Transparency note: There is no direct arbitrary-detention field in the current Core crosswalk; the same index is reused by lever 7.

LEVER 5 OF 9

Freedom from Torture and Inhumane Treatment

$2.90T global annual exclusions

Present when the state has effective measures to prevent torture. These protections represent opportunities to support human capital and cooperation. Annual cost is institutional and training: an investment toward realizing greater scale.

RTLP QUESTION

Are measures in place to prevent torture and inhumane treatment?

FIELDS AND SCORING RULE

Source: V-Dem
Field(s): v2cltort
Binary rule: v2cltort >= 2.0
Operational proxy: Freedom from torture

Transparency note: Direct substantive proxy, converted from the model estimate to a binary atlas score.

LEVER 6 OF 9

Civilian Protection in Conflict Zones

$1.36T global annual exclusions

Present when mechanisms exist to protect civilians during conflict. These protections represent opportunities to preserve the productive capacity and future of entire regions. Annual cost is primarily military discipline and rules of engagement.

RTLP QUESTION

In conflict areas, are mechanisms protecting civilians?

FIELDS AND SCORING RULE

Source: V-Dem
Field(s): v2clkill
Binary rule: v2clkill >= 2.0
Operational proxy: Freedom from political killings

Transparency note: The processed 2026 data reuse lever 3's field. The loader contains v2x_clphy, but that field did not populate this released component table.

LEVER 7 OF 9

Access to Justice

$976.7B global annual exclusions

Present when ordinary people can bring serious claims and have them heard fairly. These conditions represent opportunities for contracts and rights to function for a wider share of the population. Annual cost is court capacity and legal aid.

RTLP QUESTION

Can individuals obtain fair remedies for right-to-life violations?

FIELDS AND SCORING RULE

Source: V-Dem
Field(s): v2xcl_acjst
Binary rule: v2xcl_acjst >= 0.5
Operational proxy: Access to justice index

Transparency note: The processed 2026 data reuse lever 4's field. The loader contains v2clrspct, but that field did not populate this released component table.

LEVER 8 OF 9

Freedom of Expression and Whistleblower Protections

$965.7B global annual exclusions

Present when laws protect people who report corruption from retaliation. These conditions represent opportunities for information to flow more accurately and for public resources to be used more effectively. Annual cost is low: mainly passing and enforcing the law.

RTLP QUESTION

Are expression and whistleblower rights upheld?

FIELDS AND SCORING RULE

Source: V-Dem
Field(s): v2x_freexp
Binary rule: v2x_freexp >= 0.5
Operational proxy: Freedom of expression and alternative information index

Transparency note: Expression is the operational proxy; the current crosswalk has no direct whistleblower-law variable.

LEVER 9 OF 9

Socioeconomic Conditions

$1.72T global annual exclusions

Present when a society makes the deliberate choice to maintain a minimum floor against extreme deprivation. This reflects political will rather than just economic outcomes. When present, it increases the share of the population able to participate productively, expanding overall economic scale. Annual cost is a political decision about minimum security standards.

RTLP QUESTION

Are food and minimum material conditions adequate to sustain life?

FIELDS AND SCORING RULE

Source: World Bank WDI
Field(s): SN.ITK.DEFC.ZS, SI.POV.DDAY
Binary rule: undernourishment <= 5% AND poverty headcount <= 10%
Operational proxy: Prevalence of undernourishment and international-poverty headcount

Transparency note: Missing values are conservatively treated as failure. This is an outcome-based basic-needs proxy, not a V-Dem field.

GDP Regression, Significance, and Eta

The canonical atlas model is a population-weighted cross-sectional regression of log GDP per capita on activated-lever count. It estimates association with the level of GDP per capita, not a causal time-series growth effect.

ETA	95% ROBUST INTERVAL	MODEL FIT
0.305	8.3% to 57.3%	R2 0.252
exp(beta) - 1	n = 187	p = 0.00568

Specification: log(GDP per capita) = intercept + beta x activated-lever count + error. The fitted log slope is 0.266 with robust standard error 0.095. Eta is exp(beta)-1 = 0.305; the operational atlas rounds this to 0.30 and applies Delta G = min(eta x (1-R) x G0, 0.25 x G0).

INTERPRETATION LIMIT

Statistical significance is not causal identification. Omitted variables, reverse causation, proxy reuse, binary thresholds, and source-year mismatch remain. The 25% cap limits attribution but does not resolve those identification problems.

Regression Comparators and Growth Check

Single predictor	n	Slope	R2	p
Population	187	-0.161	0.063	0.000188
Undernourishment	159	-0.094	0.468	1.19e-14
Extreme-poverty headcount	61	-0.115	0.351	2.49e-05

Available growth falsification check: 2023-2024 current-US$ GDP-per-capita change on lever count, unweighted OLS. n=179; slope=-0.170 percentage points per lever; R2=0.004; p=0.207. This one-year check is not statistically significant. The atlas therefore describes a cross-sectional GDP-level association and does not claim that the nine levers predict one-year GDP growth.

Comparator models are descriptive single-predictor checks. Undernourishment and poverty are development outcomes and inputs to lever 9, not independent causal controls; missing data change their sample sizes.

V-Dem Dataset and Reproduction

The committed 2026 score table identifies V-Dem Country-Year Full+Others v15 as its source and uses 2024 country-year observations. The current update workflow targets Country-Year Core v16; Full+Others v16 is also accepted. The repository does not redistribute the large raw source CSV.

Official v15 archive: https://www.v-dem.net/data/dataset-archive/
Official current dataset page: https://www.v-dem.net/data/the-v-dem-dataset/
Official v16 codebook: https://www.v-dem.net/documents/70/codebook_v16.pdf

For release-vintage replication, download Full+Others v15. For a current-data update, download Core or Full+Others v16. Extract the CSV into data/raw/ with its matching versioned filename, then run python3 -m src.build_atlas --year 2026 --eta 0.30.

V-Dem's unsuffixed expert-coded fields are measurement-model point estimates on a standardized interval scale, not the original ordinal coder categories. The atlas cutoffs are project operationalizations. A v16 rebuild may revise historical scores and is therefore an update rather than guaranteed bit-for-bit replication. Exact replication requires the archived v15 source and checksum; that checksum is not present in this repository.

World Bank Dataset and Reproduction

World Development Indicators supply NY.GDP.PCAP.CD (GDP per capita, current US$), SP.POP.TOTL (population), SN.ITK.DEFC.ZS (undernourishment), and SI.POV.DDAY (international-poverty headcount). GDP is G0 and the regression outcome; population is the regression weight and aggregation factor; the last two fields populate lever 9.

World Bank Data portal: https://data.worldbank.org/
Public API: https://api.worldbank.org/v2/

The pipeline fetches the public API through wbgapi and stores processed snapshots under data/processed/.

The 2026 baseline is the freshest published value available at build time, not a claim that every value was observed in 2026. Missing undernourishment or poverty is conservatively scored as lever 9 absent. Current-US$ GDP supports current-dollar totals but also reflects price-level and exchange-rate movement.

Index of Terms

2026-fresh G0:
: The most recent published GDP per capita values used as the dynamic baseline for the 2026 edition (even when RTLP components are from 2024).

Access to Justice:
: Fair access to remedies for right-to-life violations

Arbitrary Detention:
: Core protection against state overreach (RTLP indicator #4); current implementation level contributes to associated capital exclusions in regions.

Attributable capital exclusions:
: The share of a country or region's total capital exclusions (lost potential GDP) that is directly traceable to the current implementation level of one specific protection (sum of $\eta/9 \times G_0 \times$ pop over countries where the protection is not fully present, with current $\eta \approx 0.30$ from population-weighted cross-section).

Binarization / threshold:
: The documented conversion of continuous V-Dem and World Bank values into the binary 0/1 scores for the nine indicators (e.g., ≥2.0 on 0-4 scales, ≤5 % undernourishment + ≤10 % poverty for #9).

Causality and Attraction:
: The source monograph (Hubbard 2026V3, Zenodo 19468550) that supplies the RTLDI equation, the nine RTLP indicators, the nested-enclosures paradigm, and the cartographic critique of distortion. This atlas extends the source RTLDI to quantify capital exclusions.

Choropleth:
: Thematic map in which countries or regions are shaded by enclosure strength R (0–1) using the RTLDI atlas scale; primary visual for the global, regional, and focus views.

Civilian Protection in Conflict Zones:
: Mechanisms protecting civilians in conflict areas

Country view:
: The country-only choropleth on each nation page, fitted to the national boundary and shaded by enclosure strength R without using the regional context map.

Crosswalk:
: The explicit mapping from the nine verbatim RTLP questions to observable V-Dem variables and World Bank statistics, with defined binarization thresholds.

ΔG (delta G):
: Annual per-capita capital exclusions (lost potential GDP) associated with the current level of the nine levers. Core output of the RTLDI equation.

Economic Intelligence (tool):
: The use of R scores, component breakdowns, and attributable capital exclusions figures as practical information for governments, businesses, and analysts to identify high-leverage institutional opportunities.

Enclosure (nested causal / human):
: The source document's central metaphor: the interdependent, layered structures of protection whose strength or weakness shapes higher-order economic and social steady states.

η (eta):
: The sensitivity coefficient (current data-driven default 0.30 from population-weighted 2026 UN cross-section). In the bounded model $\Delta G = \min(\eta(1 - R) \times G_0, 0.25 \times G_0)$; the cross-sectional premium per indicator, then capped so the nine RTLP indicators are never credited with more than 25% of observed G_0 after other determinants are given due weight. (Source document used a more conservative 0.05 structural value.)

Existence of Legal Protections:
: Provisions explicitly protecting the right to life

Freedom from Torture and Inhumane Treatment:
: Measures in place to prevent torture

Freedom of Expression and Whistleblower Protections:
: Expression and whistleblower rights upheld

G_0 (G-zero):
: Baseline GDP per capita (World Bank, current US$). The 'current' economic size against which capital exclusions are measured.

Independent Judiciary:
: Independent judiciary capable of upholding right-to-life laws

Law Enforcement Accountability:
: Law enforcement agencies accountable for unlawful killings

Member-nations table:
: The compact table on each regional summary page that lists every country in the region together with its R, G0, population, and capital exclusions, plus a bottom row of regional totals.

Mollweide:
Equal-area pseudocylindrical map projection used for all choropleths (global, regional, and per-nation zooms) to ensure visual area corresponds to geographic reality.

Protection Against Arbitrary Detention:
Arbitrary detention prohibited with legal recourse

R (lever activation score):
Average activation level (0–1) of the nine Right to Life Deficit Index levers. 1 = full activation across all dimensions.

Right to Life:
The overarching protected interest; the nine indicators operationalize equal and effective protection against arbitrary interruption.

RTLDI:
Right to Life Deficit Index — the full framework from the source (Zenodo 10.5281/zenodo.19468550), extended here to quantify capital exclusions (lost potential GDP where capital is excluded because the nine protections necessary for it to operate are absent).

RTLDI atlas R scale:
The shared perceptual color scale used for every enclosure-strength choropleth in the atlas: deep low-R tones through teal midrange values to warm high-R values.

Socioeconomic Conditions:
Access to healthcare, food, and shelter adequate to sustain life

Steady state:
A balanced biological, economic, or social condition that the framework argues is supported or undermined by the strength of the nine life protections and the resulting causal enclosures.

The Nine Levers:
The nine binary institutional conditions (RTLDI) that are averaged to produce R. Low-cost, high-return switches identified through modelling for maximal prosperity.

Torture and Inhumane Treatment:
RTLP indicator #5; one of the larger contributors to global capital exclusions, highlighting physical integrity protections.

Trend (3-year):
The per-nation GDP-driven RTLDI plots (R held fixed at 2024 values) that illustrate recent dynamics in capital exclusions alongside the regional and global maps.

UN region / regional summary:
One of the 22 geographic groupings used for aggregates, focused choropleths, member-nations tables (with R / G0 / population / capital exclusions + REGIONAL TOTAL row), and 9-lever breakdowns.

V-Dem:
Varieties of Democracy project. Supplies the expert-coded governance and civil liberties indicators for eight of the nine RTLP components.

Credits and Acknowledgments

Conceptual Framework and Equations
Sid J.A. Hubbard — author of Causality and Attraction: A Continuum of Steady States (v3, 2026). All rights to the RTLDI equations, the nine RTLP indicators, and the nested causal paradigm remain with the author.

Data
• V-Dem Institute (https://www.v-dem.net) — published scores use Country-Year Full+Others v15, 2024 observation year; current update workflow targets v16.
• World Bank Open Data (https://data.worldbank.org) — WDI via API.
• United Nations — list of 193 Member States.

Code and Atlas Production
RTLDI ATLAS project (2026). The open-source toolkit, crosswalk implementation, data pipeline, and this PDF generator are provided to make the framework usable by NGOs and researchers.

This work is released in the spirit of the original document: to map, with transparent data and reproducible math, the economic consequences of failing to protect the right to life equally.

For the full source code, 2026 data tables, and previous editions (2023, 2024), see the project repository. Users are encouraged to download the latest V-Dem and World Bank releases and re-run the pipeline for updated figures.